EXP NO: 13 **Experiment to handle missing values and inappropriate data in a given data set**

AIM:

To identify and handle missing values and inappropriate data in a given dataset using Python libraries such as Pandas and NumPy by applying suitable data cleaning techniques, including removing, replacing, and correcting invalid values.

PROGRAM:

```
import numpy as np

import pandas as pd

from google.colab import files


uploaded = files.upload()

df = pd.read_csv("Hotel_Dataset.csv")

df


df.duplicated()


df.info()


df.drop_duplicates(inplace=True)

df


len(df)


df.reset_index(drop=True, inplace=True)

df


df.drop(["Age_Group.1"], axis=1, inplace=True)

df
```

```
df.loc[df["CustomerID"] < 0, "CustomerID"] = np.nan
df.loc[df["Bill"] < 0, "Bill"] = np.nan
df.loc[df["EstimatedSalary"] < 0, "EstimatedSalary"] = np.nan

df.loc[(df["NoOfPax"] < 1) | (df["NoOfPax"] > 20), "NoOfPax"] = np.nan
df

df["Age_Group"].unique()

df["Hotel"].unique()

df["Hotel"] = df["Hotel"].replace("Ibys", "Ibis")
df

df["FoodPreference"].unique()

df["FoodPreference"] = df["FoodPreference"].replace(
    ["Vegetarian", "veg"], "Veg"
)
df["FoodPreference"] = df["FoodPreference"].replace(
    "non-Veg", "Non-Veg"
)
df

df["EstimatedSalary"] = df["EstimatedSalary"].fillna(
    round(df["EstimatedSalary"].mean())
)
```

```
df["NoOfPax"] = df["NoOfPax"].fillna(
    round(df["NoOfPax"].median())
)

df["Rating(1-5)"] = df["Rating(1-5)"].fillna(
    round(df["Rating(1-5)"].median())
)

df["Bill"] = df["Bill"].fillna(
    round(df["Bill"].mean())
)

df

df.info()
print("Total rows:", len(df))
print("Total columns:", len(df.columns))
print("Missing values:")
print(df.isnull().sum())
```

OUTPUT:

```
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       11 non-null     int64
 1   Age_Group        11 non-null     object
 2   Age_Group.1      11 non-null     object
 3   Hotel            11 non-null     object
 4   FoodPreference   11 non-null     object
 5   Bill             10 non-null     float64
 6   EstimatedSalary  10 non-null     float64
 7   NoOfPax          10 non-null     float64
 8   Rating(1-5)      10 non-null     float64
dtypes: float64(4), int64(1), object(4)
memory usage: 924.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 8 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   CustomerID       9 non-null      float64
 1   Age_Group        10 non-null     object
 2   Hotel            10 non-null     object
 3   FoodPreference   10 non-null     object
 4   Bill             10 non-null     float64
 5   EstimatedSalary  10 non-null     float64
 6   NoOfPax          10 non-null     float64
 7   Rating(1-5)      10 non-null     float64
dtypes: float64(5), object(3)
memory usage: 772.0+ bytes
Total rows: 10
Total columns: 8
Missing values:
CustomerID         1
Age_Group          0
Hotel              0
FoodPreference     0
Bill               0
EstimatedSalary    0
NoOfPax            0
Rating(1-5)        0
dtype: int64
```

RESULT:

Thus, the missing values and inappropriate data in the given dataset were successfully identified and handled using Python Pandas, and the dataset was cleaned for further analysis.